19 · Static and dynamic equilibrium

The words of this track, as it says them. Bold brown is the examiner.

Static versus dynamic equilibrium. Four marks.

What is the difference between static and dynamic equilibrium? How, and where, is each one detected?

The hook: static is gravity and straight lines, in the vestibule, where little stones slide. Dynamic is spinning, in the semicircular canals, where fluid swings a gel cap.

What they share. Both are sensed in the vestibular apparatus of the inner ear, by hair cells, which are mechanoreceptors. The hair cells fire when fluid, the endolymph, or a jelly mass moves and bends their hairs.

Static equilibrium is the position of your head in response to gravity, and to straight-line movement. Tilting your head. Moving forwards or backwards. Going up in a lift. It's detected by the maculae, in the utricle and the saccule, which together make up the vestibule. The hairs there are embedded in a jelly otolith membrane, studded with tiny stones called otoliths. When you tilt your head, the stones slide and bend the hairs.

Dynamic equilibrium is balance while your head is rotating: turning, or spinning. It's detected by the crista ampullaris, in the ampulla at the base of each semicircular canal. Rotation moves the endolymph, which swings a gel cap called the cupula, and that bends the hairs.

Both send impulses along the vestibular branch of cranial nerve eight, to the brain stem and the cerebellum. And signals that fire with no real movement cause vertigo.

The traps. Static is the vestibule: the utricle, the saccule, the maculae, the otoliths. Dynamic is the semicircular canals: the ampulla, the crista, the cupula. Swap them and you lose both marks. Second: static doesn't mean no movement at all. A lift going up is static, because it's a straight line. Dynamic means rotation. Third: the cochlea is for hearing, not balance.

Once more. Static and dynamic equilibrium.

Both are sensed by hair cells in the vestibular apparatus of the inner ear, and sent along nerve eight to the brain stem and cerebellum. Static equilibrium is head position relative to gravity, and straight-line movement. It's detected by the maculae in the utricle and saccule, where otoliths slide when the head tilts. Dynamic equilibrium is balance during rotation. It's detected by the crista ampullaris in each semicircular canal, where the endolymph swings the cupula and bends the hairs.